

Property 1: Commutativity

To Prove: $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ for all $\mathbf{x}, \mathbf{y} \in \mathbf{V}$

Proof: Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ be vectors in $\mathbf{V}$.

By the definition of vector addition:

- $\mathbf{x} + \mathbf{y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$
- $\mathbf{y} + \mathbf{x} = (y_1 + x_1, y_2 + x_2, \dots, y_n + x_n)$

Since addition in the underlying field $\mathbf{F}$ is commutative, we have:

- $x_1 + y_1 = y_1 + x_1$
- $x_2 + y_2 = y_2 + x_2$
- $\vdots$
- $x_n + y_n = y_n + x_n$

Therefore: $\mathbf{x} + \mathbf{y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) = (y_1 + x_1, y_2 + x_2, \dots, y_n + x_n) = \mathbf{y} + \mathbf{x}$

Property 2: Associativity

To Prove: $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbf{V}$

Proof: Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$, $\mathbf{y} = (y_1, y_2, \dots, y_n)$, and $\mathbf{z} = (z_1, z_2, \dots, z_n)$.

Computing the left side: $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) + (z_1, z_2, \dots, z_n) = ((x_1 + y_1) + z_1, (x_2 + y_2) + z_2, \dots, (x_n + y_n) + z_n)$

Computing the right side: $\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (x_1, x_2, \dots, x_n) + (y_1 + z_1, y_2 + z_2, \dots, y_n + z_n) = (x_1 + (y_1 + z_1), x_2 + (y_2 + z_2), \dots, x_n + (y_n + z_n))$

Since addition in $\mathbf{F}$ is associative, for each component i : $(x_i + y_i) + z_i = x_i + (y_i + z_i)$

Therefore: $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$

Property 3: Identity Element

To Prove: $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbf{V}$

Proof: Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{0} = (0, 0, \dots, 0)$ be the zero vector.

Computing $\mathbf{x} + \mathbf{0}$: $\mathbf{x} + \mathbf{0} = (x_1, x_2, \dots, x_n) + (0, 0, \dots, 0) = (x_1 + 0, x_2 + 0, \dots, x_n + 0) = (x_1, x_2, \dots, x_n)$ [since $a + 0 = a$ in $\mathbf{F}$] $= \mathbf{x}$

By commutativity (Property 1): $\mathbf{0} + \mathbf{x} = \mathbf{x} + \mathbf{0} = \mathbf{x}$

Therefore: $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$

Property 4: Inverse Element

To Prove: $\mathbf{x} - \mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbf{V}$

Proof: Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and note that $\mathbf{x} - \mathbf{x} = \mathbf{x} + (-\mathbf{x})$.

The additive inverse $-\mathbf{x} = (-x_1, -x_2, \dots, -x_n)$.

Computing $\mathbf{x} + (-\mathbf{x})$: $\mathbf{x} + (-\mathbf{x}) = (x_1, x_2, \dots, x_n) + (-x_1, -x_2, \dots, -x_n) = (x_1 + (-x_1), x_2 + (-x_2), \dots, x_n + (-x_n)) = (0, 0, \dots, 0)$ [since $a + (-a) = 0$ in $\mathbf{F}$] $= \mathbf{0}$

Therefore: $\mathbf{x} - \mathbf{x} = \mathbf{0}$